ChemicalQDevice | Independent Research Paper | Draft 1.0

10 Funding Applications

A Phase 1, First-in-Human, PDAC Clinical Trial Protocol of a LLM-Directed Robotic Whipple with Daraxonrasib (RMC-6236)

Draft 1.0 | Repository v4.4.0 | 20 figures | 10 recipients

01 NIH Pioneer Award	02 ARPA-H	03 NSF TIP X-Labs	04 DOE Genesis Mission	05 NIH SEED SBIR
06 FNIH AMP	07 HHMI Investigator	08 NCI CTEP	09 Convergent FRO	10 UC San Diego Moores

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Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

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The paper (v1.0) is at `https://doi.org/10.5281/zenodo.xxxxxxxx` and the repository (v4.4.0) is at `https://github.com/kevinkawchak/physical-ai-oncology-trials`. The ten application file sets summarised here are in funding/pdac-funding-applications.

Keywords Independent scientist; Federal research portfolio; Pancreatic ductal adenocarcinoma; Robotic pancreaticoduodenectomy; Daraxonrasib; Physical AI; On-premises LLM; Phase 1 trial; Funding mechanisms

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Abstract

In July 2026 the White House Office of Science and Technology Policy asked the federal research portfolio, approximately $200 billion annually, to prioritise the individual scientist over legacy institutions [1]. This paper reports what happens when one unaffiliated scientist takes that instruction literally and writes ten funding applications for the same Phase 1 pancreatic cancer trial, each to a different recipient, each addressing the mechanism that recipient runs.

The trial is constant across all ten: an open-label, single-arm, 3+3 dose escalation of perioperative daraxonrasib (RMC-6236) in up to 18 participants with resectable or borderline-resectable KRAS G12-mutated pancreatic ductal adenocarcinoma, performed as a staged eight-arm robotic pancreaticoduodenectomy with an on-premises large language model confined to advisory output [2, 3]. Five applications lead with the operation and five with the drug and patient selection; both sets describe the same hybrid procedure. UC San Diego Moores Cancer Center is named in all ten as the intended partner of choice, approached at the feasibility stage only, and is described nowhere as a partner, sponsor, trial site, or endorser.

The evidence base is stated with its limits attached. An August 2025 quantitative systems pharmacology simulation returned median overall survival of 12.8 months for a daraxonrasib combination against 5.4 months for chemotherapy [4]; in May 2026 the RASolute 302 trial reported 13.2 against 6.6 months in the RAS G12 metastatic population [5]. That proximity is a chronology observation, not a validation claim, and the three differences are stated wherever it appears. Neither result speaks to the resectable setting, which is what the proposed trial would measure.

The applications themselves carry no digital object identifiers. This paper carries one.

How to Read This Paper

This is a summary of ten application file sets, not an application. It reports what was written, to whom, and on what evidence, and it is written so that a reader can check every claim against a named file in a public repository [6].

It is not medical advice, not regulatory advice, and not a claim that any institution, sponsor, or drug developer has agreed to anything. No patient has been treated. The trial does not exist until a site, an institutional review board, an active investigational new drug application, monitoring, insurance, and a sponsor of record are all in place.

Three reading paths. A **funder** should read §2 for the recipient set and §8 for the budget in its ten shapes. A **clinician** should read §6 for the governance boundaries and §5 for the evidence with its limitations. A **meta-science reader** should read §1 for the policy argument and §9 for the build method, which is reproducible from the prompt and the sources alone.

Figure and Table Inventory

Twenty figures across five machine-readable diagram platforms. The split follows what each figure has to answer rather than an equal quota: an even division across five platforms would have forced at least four figures into a vocabulary that cannot state their subject.

Fig	Type	Sec	The question it answers
1	mermaid	§1	How does one policy sentence become ten addressed applications?
2	plantuml	§1	Where does an unaffiliated proposal stall, and what removes each stall?
3	d2	§2	How do the ten compare on mechanism, term, ask, and lead?
4	mermaid	§5	How long did each activity run, and what overlapped?
5	plantuml	§3	Who is authorised to take which action?
6	graphviz	§2	Which award unblocks which activity, and what is redundant?
7	d2	§5	Which evidence tier contains which, and what may not be cited as what?
8	mermaid	§3	What are the reviewer's gates, and which section answers each?
9	diagrams	§6	What runs inside the trust boundary, and what may cross it?
10	graphviz	§6	By what path could an unintended motion reach the patient?
11	d2	§8	Where does each dollar land, and what is contributed rather than paid?
12	mermaid	§6	Who says what to whom across one operative day?
13	plantuml	§6	Under what guard does the advisory system change state?
14	diagrams	§6	Which model produced which artifact, and who reviewed it?
15	d2	§6	What joins to what in the released dataset?
16	graphviz	§5	Which prior work fed which, and where do the two lines meet?
17	mermaid	§8	How long is each review clock against the site milestone?
18	diagrams	§9	How does one operator's stack compare with a program office?
19	plantuml	§4	What runs in parallel once any one award lands?
20	graphviz	§10	Which field can falsify the programme's central claim?

The twenty figures, their platform, their section, and the question each exists to answer. Six are mermaid-type, four d2-type, four graphviz-type, three plantuml-type, and three diagrams-python-type.

Table	Sec	Contents	Source directory
1 to 2	§0	Figure and table inventories	The five stage directories
3 to 4	§1	Where the enterprise stalls; the four goals answered	`funding/science-golden-age/`
5 to 6	§2	The ten recipients; what differs and what does not	`applications/`
7	§3	Set A figure inventory by type	`applications/app-01` to `app-05`
8	§4	Consortium measurements against resection timing	`applications/app-06`
9 to 11	§5	Results with stated limitations; cost and schedule; the fourteen works	`funding/supplementary/source-files/`
12	§6	The three frontier-model roles	`funding/tripartisan-llm-support.md`
13 to 14	§7	Three positioning corrections; two site routes	`funding/potential-partners/`
15	§8	One budget in ten shapes	`funding/RFA-RM-27-001-v2/`
16 to 17	§9	The thirteen sub-prompts; the diagram split	`sub-prompts/`
18 to 19	§10	What is not claimed; risks the programme cannot retire alone	Application back matter

The paper's nineteen tables, the section each appears in, and the repository directory its values are drawn from. No value in any table is computed here for the first time.

1 The $200 Billion Realignment and the Individual Scientist

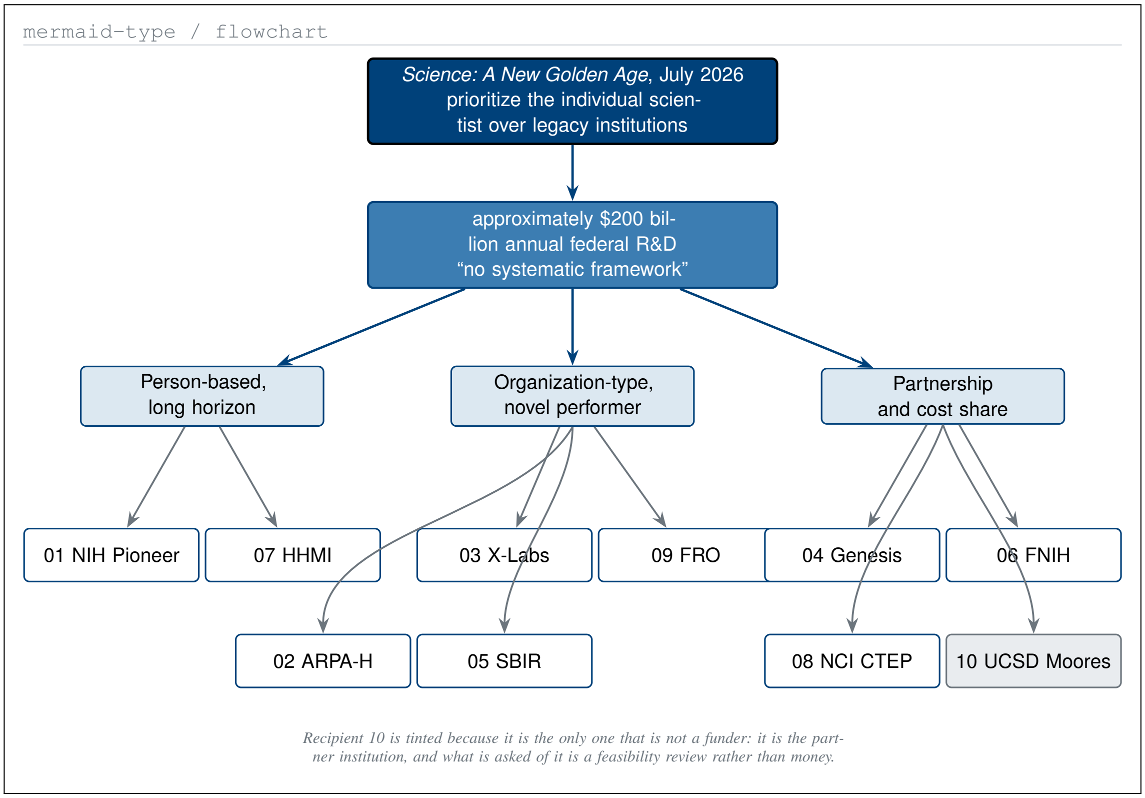
1.1 What the Report Actually Says

The July 2026 report to the President, *Science: A New Golden Age*, states four overarching goals in its letter of transmittal [1]. The first is that the United States research system should prioritise the individual scientist over legacy institutions. The second is that the way research dollars are allocated, distributed, and assessed should fundamentally change. The third is that clear scientific goals should be set and industrial capacity built to translate discovery into technological strength. The fourth is that the research enterprise should be prepared for the AI revolution.

The first goal is stated without hedging: "If we are serious about expanding what our scientists can do, we must invest directly in American researchers and the bold ideas that drive them. Too much of our research enterprise has come to serve itself rather than the scientists within it." The Summary of the Report turns that into an instruction with a named model attached: agencies should "bet on people, not just projects," by expanding portable graduate fellowships, backing early-career independence, and "scaling long-horizon grants for the best and brightest modeled on National Institutes of Health (NIH) Director's Pioneer Award."

Chapter II attaches a number. Federal agencies distribute approximately **$200 billion** in annual research and development funding, and the report's finding about that portfolio is the sentence this paper is written against: there is "no systematic framework for identifying where those dollars could catalyze the greatest scientific returns, with deference instead given to the same incumbents that consume the funding. This process produces a portfolio that emerges by accident rather than intentional design."

The corrective the report names has three parts: realign the portfolio toward excellence and risk taking, strip away the administrative burden that consumes close to half of a federally funded investigator's working



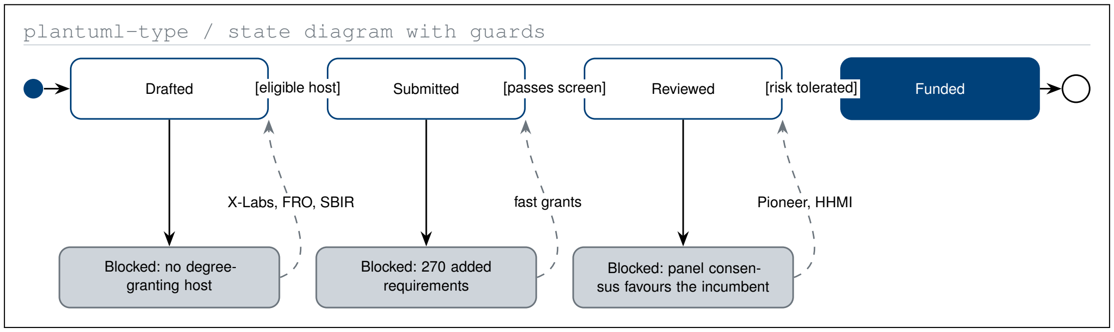
One sentence of federal policy, three mechanism families, and the ten recipients each family reaches. The split is by mechanism, not by agency, which is why one recipient is not a federal funder.

hours, and embed continuous evidence-based improvement across the whole portfolio. What the report does not do is say who should receive the realigned money. This paper is one answer, submitted ten times.

1.2 Where the Enterprise Stalls

The report's diagnosis is quantitative, and every number below is the report's own or is drawn from the primary source its end notes name.

Finding	Value	Mechanism the report names against it
Research time lost to administration	near one half	Burden reduction; applications of a few pages
New federal requirements on grants, 1991 to January 2025	at least 270	Requirement rescission; metascience units
Negotiated indirect cost rates	50 to 60 percent	Indirect-cost reform; private funders accept 10 to 15
Grant lead time	up to 20 months	Fast grants decided in under one month
NIH principal investigator average age, 1980 to 2008	39 rising to 51	Early-career independence; portable fellowships
Fall in new drugs per billion dollars since 1950	roughly eightyfold	Portfolio construction across exploration and exploitation [7]
Publication-rate advantage of person-based funding	nearly double	HHMI-style support, roughly $10M over seven years [8]



The five states a proposal from an unaffiliated scientist passes through, and the three at which the report says the enterprise currently stalls. Each stall names the mechanism that removes it.

The report's diagnosis in seven numbers, each paired with the mechanism it names as the corrective. The last row is the only one that reports an outcome rather than a friction.

An unaffiliated scientist meets a subset of these frictions and one that the table does not contain: the eligibility screen. Most federal mechanisms require a host institution before a proposal can be reviewed at all, which means the merit question is never reached. Figure 2 states the resulting state machine and names, for each stall, the mechanism the report offers against it.

1.3 Why an Independent Scientist Is the Test Case

Between June 2025 and July 2026, one person produced fourteen deposited works: two clinical protocols, an investigational new drug application, two funding applications, five legislative drafts, and four guidance works [9]. None of it carried indirect cost. Three quantitative trial simulations were completed at an estimated $36,330 each against an industry benchmark that runs from $120,000 for a single empirical virtual trial to more than $2 million for a ten-arm quantitative systems pharmacology trial [10].

The claim is a ratio and nothing more. It is not that the work is better than an institution's, and it is not that institutions are unnecessary. It is that the part of the pipeline which used to require a program office no longer does, and that the part which still requires an institution, patients and an operating theatre and an institutional review board, is exactly the part these ten applications are asking to reach.

Transmittal-letter goal	What the ten applications supply against it
Prioritise the individual scientist over legacy institutions	One named applicant, no host institution, fourteen deposited works before any award
Change how research dollars are allocated and assessed	Ten mechanism families addressed on the same science, so the mechanism is the only variable
Set clear goals and build translational capacity	A published protocol and IND, and a governance package designed to be site-portable
Prepare the enterprise for the AI revolution	An advisory system whose boundary is measured in milliseconds and released as an open schema

The report's four overarching goals, and the specific thing the application set supplies against each. The second row is what makes the set a comparison rather than ten separate requests.

d2-type / scored grid

Recipient	Mechanism	Term	Direct ask	Lead
01 NIH Pioneer Award	person-based	5 years	$3.5M	surgical
02 ARPA-H	milestone	36 months	$2.1M	surgical
03 NSF TIP X-Labs	organization	5 years	$3.5M	surgical
04 DOE Genesis Mission	mission	5 years	$3.5M	surgical
05 NIH SEED SBIR	small business	9 + 24 mo	$1.6M	surgical
06 FNIH AMP	consortium	5 years	$3.5M	medical
07 HHMI Investigator	person-based	7 years	$0.7M/yr	medical
08 NCI CTEP	concept review	5 years	$3.5M	medical
09 Convergent FRO	time-bound	5 years	$3.5M	medical
10 UC San Diego Moores	institutional	one meeting	none	medical

The science column is omitted because it would be constant: the same population, the same 3+3 escalation, the same co-primary endpoints, and the same advisory boundary appear in all ten.

Ten applications on four attributes. The science column is constant by construction: the same trial is described ten times, and only the mechanism, the term, and the ask change.

2 The Ten Applications

2.1 Selection Rule

A recipient qualifies for this set only if *Science: A New Golden Age* either names its programme or names the mechanism that programme runs. Nothing was included on plausibility. The rule matters because it is what makes the set an argument rather than a mailing list: every application can open by quoting the recipient's own mechanism back to it, in the administration's wording, and then answer it.

Three mechanism families emerged from applying that rule. **Person-based**, where the report names the NIH Director's Pioneer Award by title and cites HHMI-style support as the evidence. **Organization-type**, where the report names X-Labs as the first federal programme designed to fund independent research organizations outside traditional academia, names focused research organizations as time-bound nonprofits built to dissolve, names the ARPA program-manager model, and names SBIR as opening doors for technician-founded ventures. **Partnership and cost share**, where the report holds up the Foundation for the NIH and the Accelerating Medicines Partnership as its model, names the Genesis Mission and its Robotics line, and asks agencies to prioritise non-federal cost share [11].

2.2 What Differs, and What Does Not

No	Recipient	Mechanism family	The anchor sentence in the report
01	NIH Common Fund, Director's Pioneer Award	Person-based	"scaling long-horizon grants for the best and brightest modeled on . . . Director's Pioneer Award"
02	ARPA-H mission office	Organization-type	The ARPA program-manager model, with ARPA-H and ARPA-E as the live examples
03	NSF TIP Directorate, X-Labs	Organization-type	"the first federal program designed to fund independent research organizations outside traditional academia"
04	DOE Office of Science, Genesis Mission	Partnership	Robotics mission: "to initiate the era of physical AI-driven scientific discovery"
05	NIH SEED, SBIR and STTR	Organization-type	"Programs like SBIR open doors for technician-founded ventures"
06	Foundation for the NIH, AMP	Partnership	FNIH and the Accelerating Medicines Partnership held up as the model
07	HHMI Investigator Program	Person-based	Person-based funding of roughly $10 million over seven years
08	NCI Cancer Therapy Evaluation Program	Partnership	Prioritised foundational biological sciences; the report's own cancer framing
09	Convergent Research, FRO programme	Organization-type	Focused research organizations as time-bound nonprofit research startups
10	UC San Diego Moores Cancer Center	Partnership	Regional innovation clusters; non-federal cost share

The ten recipients with the sentence in the report that qualifies each. No recipient appears without an anchor, which is what makes the set an argument inside the funder's own frame rather than a mailing list.

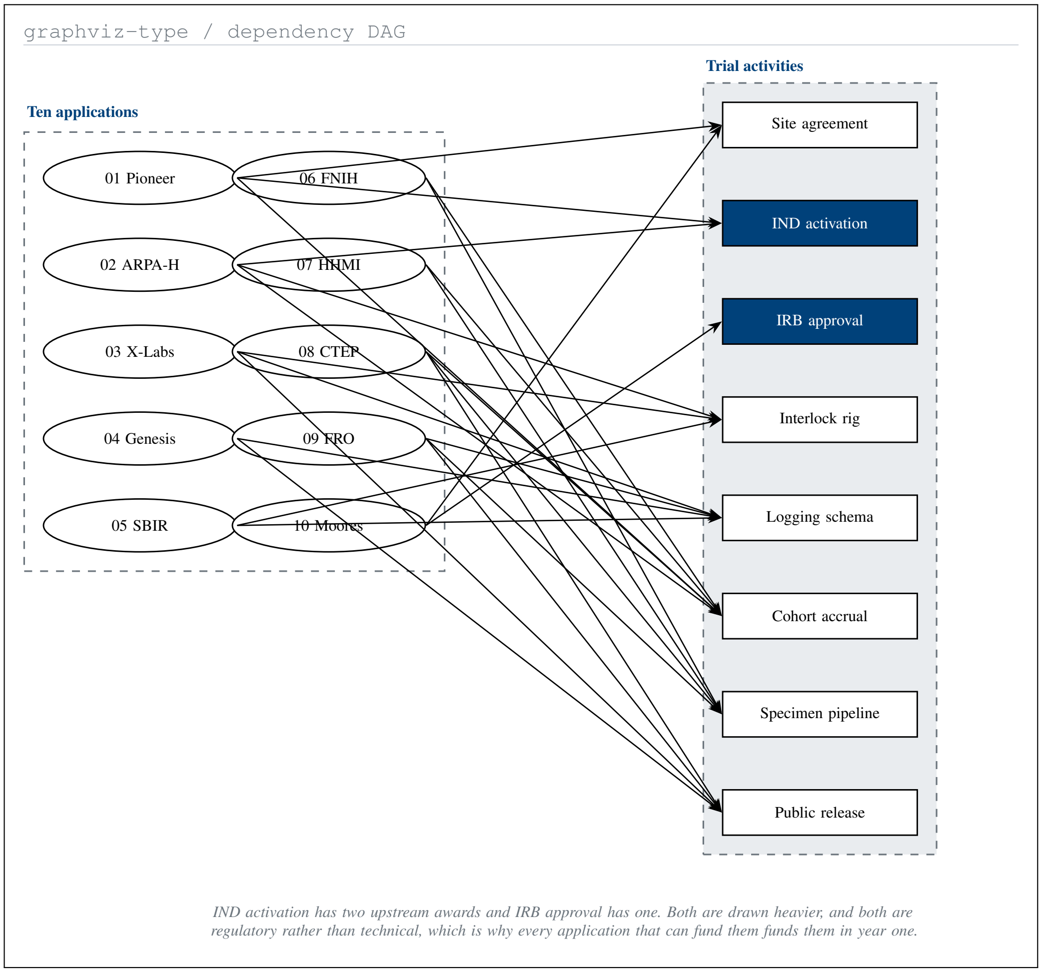
What is constant is the trial. Every one of the ten describes the same population, resectable or borderline-resectable KRAS G12-mutated pancreatic ductal adenocarcinoma; the same design, open-label, single-arm, 3+3 escalation, up to 18 treated participants; the same co-primary endpoints; and the same advisory boundary, with arm-level stop specified at 3 milliseconds and system-wide stop at 500 milliseconds [2, 3].

That constancy is the point. A reviewer who compares two applications from this set is comparing two funding mechanisms applied to one piece of science, with the science held fixed. It is the closest thing an individual applicant can construct to a controlled comparison of mechanisms, and it is only possible because the protocol was published before the applications were written.

2.3 Redundancy Across the Set

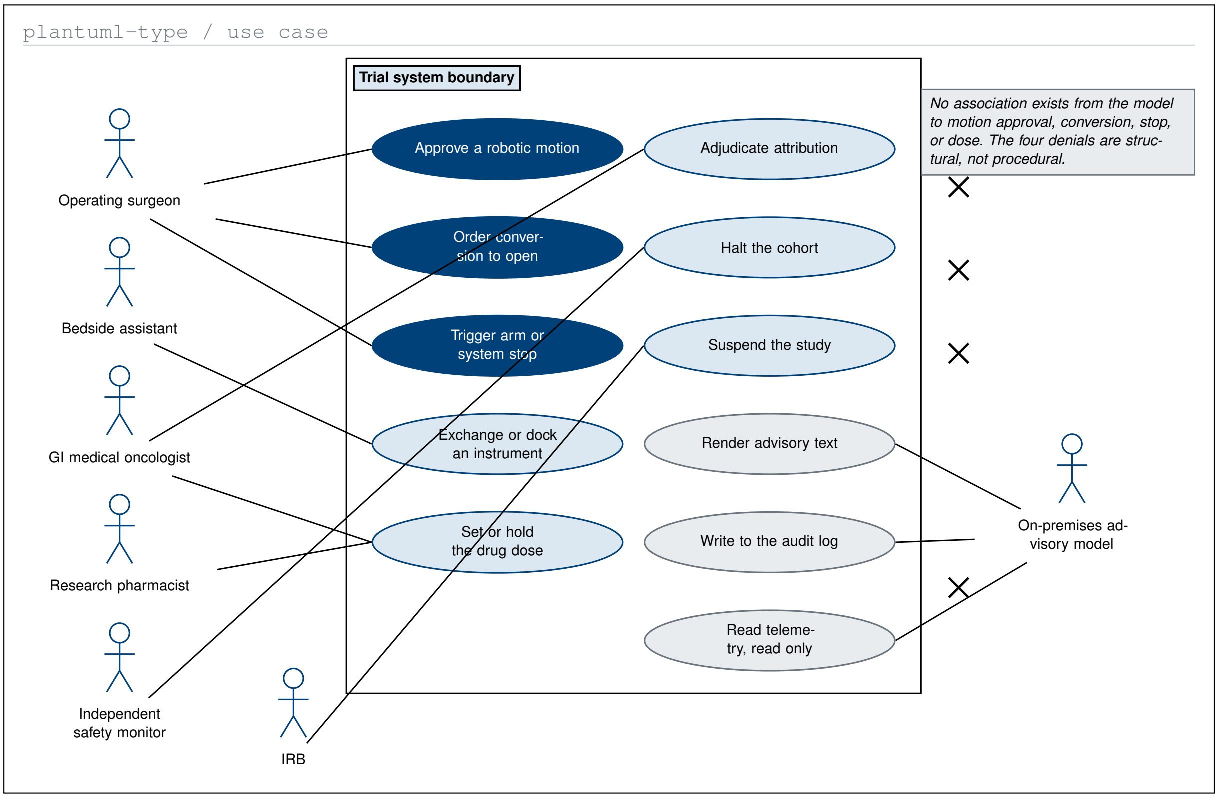
Sending ten applications is not ten times one application. Because the ten fund overlapping activity sets, most of the programme survives a refusal from any one recipient. Figure 6 makes that concrete, and identifies the two activities that do not.

Six of the eight activities have two or more upstream awards. The interlock rig is funded by applications 02, 03, and 05; the logging schema by 03, 04, 05, and 09; cohort accrual by 01, 02, 06, 07, and 08. Any single refusal in that set leaves the activity funded.



Ten awards against eight trial activities. Six activities have two or more upstream awards and survive any single refusal; the two that do not are drawn heavier, and both are regulatory.

The two exceptions are regulatory. IND activation is reachable only through applications 01 and 02, and IRB approval only through application 10, which asks for no money at all. That asymmetry is the programme's real structure: the scarce resource is not funding but institutional standing, and the one application that could supply it is the one with nothing to offer in return.



Seven actors and the eleven actions each may take. Every action has at least one human association; the advisory model has exactly one, and the struck links are the four it is denied by construction.

3 Set A: The Surgical Perspective

3.1 What Leads in Set A

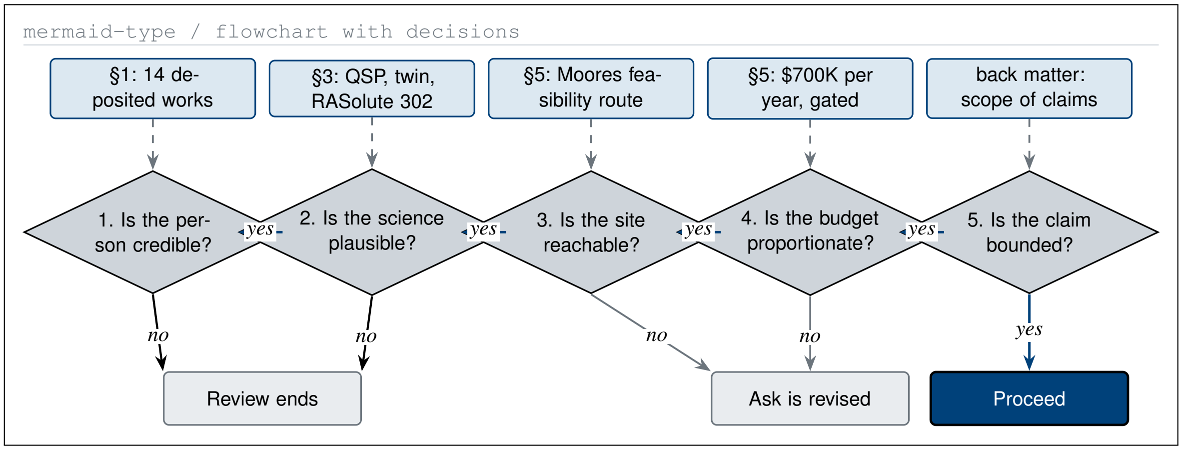
In the five surgical applications the operation leads. Daraxonrasib is present in all five, but it enters as the perioperative arm the operation is timed around rather than as the headline, and the endpoint table is led by conversion, anastomotic integrity, operative time, estimated blood loss, R0 margin status, and 90-day morbidity graded to the international definition [12].

That ordering is not cosmetic. A reviewer at a mechanism office asks whether the hard part is buildable; a reviewer at a clinical programme asks whether the hard part is survivable. Set A answers the first, and the hard part it names is the operation with a language model in the room.

3.2 Who Is Authorised to Do What

The single most common objection to a language model in an operating theatre is that the boundary is asserted rather than specified. Applications 02 and 10 each carry a three-actor slice of the authority question. Figure 5 gives the whole set, including the two actors that appear in no application figure: the research pharmacist and the institutional review board.

Three fills separate three authority classes. Corporate Blue marks the three actions only the operating surgeon takes. Pale blue marks the shared clinical and oversight actions. Gray marks the three actions the model may take, all of which produce text or read data and none of which moves anything. A reader who ignores the labels entirely still reads the structure correctly.



The five questions a reviewer asks in order, and the section of the application that answers each. A no at gate 1 or gate 2 ends the review; a no at gates 3 to 5 changes the ask rather than ending it.

3.3 Applications 01 to 05

01, NIH Director's Pioneer Award. The unit of risk is the individual rather than the aim. The bet, that a targeted RAS(ON) inhibitor and a robotic operation can be evaluated as one governed system, does not survive decomposition into three independent project aims, because the drug schedule, the operative plan, and the advisory boundary are only interesting together. The ask is $700,000 per year for five years, with a stated go or no-go condition at each of three gates.

02, ARPA-H. Written as a gated programme rather than a project narrative, because the report endorses the ARPA model on the specific ground that a program manager can end a bet. Three gates at months 12, 21, and 33, each carrying a numeric kill criterion checkable from programme data: IND not active or measured stop latency above specification; any unexpected grade 4 event attributed to the device or the advisory path; fewer than two evaluable participants at any dose level.

03, NSF TIP X-Labs. Argues organization type rather than money. The programme needs five activities held together, operating under an active IND, maintaining a verification harness, publishing negative operative results, holding the regulatory package as a maintained product, and releasing code and data openly. No existing performer type covers all five, and that gap is what the report says X-Labs was created to fill.

04, DOE Genesis Mission. The FY 2028 annex defines the Robotics national mission as "general-purpose autonomous systems capable of dexterous manipulation, mobility, and reliable operation in real-world environments, to initiate the era of physical AI-driven scientific discovery" [11]. A human operating room is the hardest available instance of that sentence. Three artifacts are offered to the mission rather than to the clinic: a trust-boundary topology, a closed loop with one measured human gate, and open data formats [13].

05, NIH SEED SBIR. The only application in which the commercial product carries the argument. The deliverable is not the robot and not the drug but the governance package that makes an on-premises model usable in a regulated theatre: a logging schema, a bench stop-latency method, and a regulatory dossier a qualified programme can adopt without rebuilding either.

3.4 What a Reviewer Actually Asks

The asymmetry between the first two gates and the last three is why the applications are ordered as they are. Credibility and plausibility are settled in the first two sections of every attachment, before any ask appears, because a reviewer who is unconvinced there never reaches the budget.

No	Figures	Types used	Type deliberately not used, and why
01	3	mermaid, d2, graphviz	No plantuml or diagrams: no authority question, no topology question
02	4	mermaid, plantuml, d2, graphviz	No diagrams: nothing is deployed across a boundary in this argument
03	3	d2, diagrams, mermaid	No plantuml or graphviz: the argument is organization type, not authority or dependency
04	4	diagrams, mermaid, graphviz, d2	No plantuml: the guards are stated in prose and drawn in the summary paper instead
05	3	mermaid, graphviz, d2	No plantuml or diagrams: a phased commercial argument needs neither

Figure inventory for Set A, including the platform each application deliberately did not use. Recording the omission is what keeps the type choice a decision rather than a default.

4 Set B: The Medical Oncology Perspective

4.1 What Leads in Set B

In the five medical oncology applications the drug and the patient selection lead. Daraxonrasib (RMC-6236) is a RAS(ON) multi-selective inhibitor, given perioperatively in KRAS G12-mutated resectable or borderline-resectable disease, and the operation is described in full but as the instrumented setting that makes the pharmacology measurable.

The endpoint table reorders accordingly: dose-limiting toxicity and the 3+3 escalation first, then pharmacokinetic and pharmacodynamic sampling timed around the resection, then pathologic response, ctDNA clearance, and 24-month overall survival. The operation matters to the pharmacology for one specific reason, which no metastatic study can supply: it is the only setting in which paired pre-treatment and on-treatment human tumour tissue can be obtained under controlled timing.

Measurement	Timing relative to resection	Why no other setting supplies it
Plasma pharmacokinetics	Days -14, -1, and $+1$	Exposure in a surgical population has not been characterised by any sponsor
Tumour pathway pharmacodynamics	Resection specimen, day 0	Cannot be obtained in the metastatic setting at all
ctDNA clearance kinetics	Days -14, $+30$, $+90$	Defines a shared surrogate anchored to a known resection date
Pathologic response grade	Resection specimen, day 0	Anchors every later perioperative RAS study to one scale

The four measurements the consortium in application 06 exists to make, when each is taken relative to the resection, and the reason a metastatic trial cannot supply any of them.

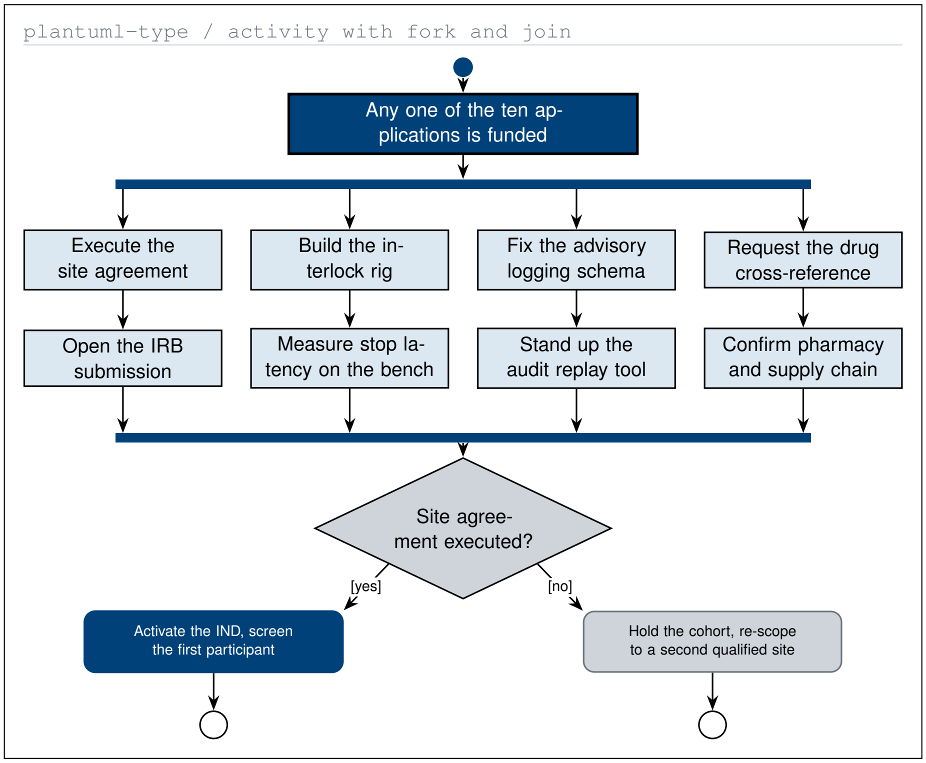
4.2 Applications 06 to 10

06, Foundation for the NIH. Proposes a narrow pre-competitive question that no single party owns: what happens pharmacologically when a RAS(ON) inhibitor is given around a curative-intent pancreatic resection. Three parties contribute, an independent scientist, an academic cancer center, and a drug developer, and the single shared output, a specimen-level pharmacodynamic dataset, is released openly and owned by none of them. That is what makes it pre-competitive: it differentiates no product.

07, HHMI Investigator Program. Tests the one place the report's person-based argument might not reach, an investigator with no academic appointment, and asks the eligibility question directly rather than burying it. The programme is written as a state machine rather than a set of aims, and its defining property is that every failure transition re-enters the programme: if exposure is inadequate the schedule changes and the agent is retained; if pathway suppression is not demonstrated the agent changes and the question is retained.

08, NCI Cancer Therapy Evaluation Program. The most clinically technical of the ten, because a concept review is a technical review. It puts one design point forward: a combined drug and device study in a surgical population carries a hazard a standard escalation does not, in that an adverse event may be attributable to the drug, to the operation, or to the advisory system, and the dose decision depends on which. Attribution is therefore adjudicated by a blinded reviewer before the toxicity window closes, and only the drug branch reaches the dose decision.

09, Convergent Research. The only application whose organization states its own end date. Three properties are put forward to be tested: the organization closes in year five and the budget contains no year six; three releases are scheduled of which only the specimen dataset depends on the trial succeeding; and every



One award, four parallel workstreams, one join. The join condition is the site agreement, not the money, which is why nine of the ten asks can be satisfied by any single award rather than by all ten.

artifact has one named recipient class fixed in year one, because negotiating recipients in year five is how a dissolved organization strands its own outputs.

10, UC San Diego Moores Cancer Center. Different in kind. It is addressed to the partner institution rather than to a funder, asks for a 45-minute feasibility meeting rather than money, and carries three positioning corrections drawn from the author's own site research: daraxonrasib is not first-in-human; the robotic configuration must be named by manufacturer, model, cleared configuration, arm count, instruments, and software version before feasibility can be assessed; and no drug supply or letter of authorization exists.

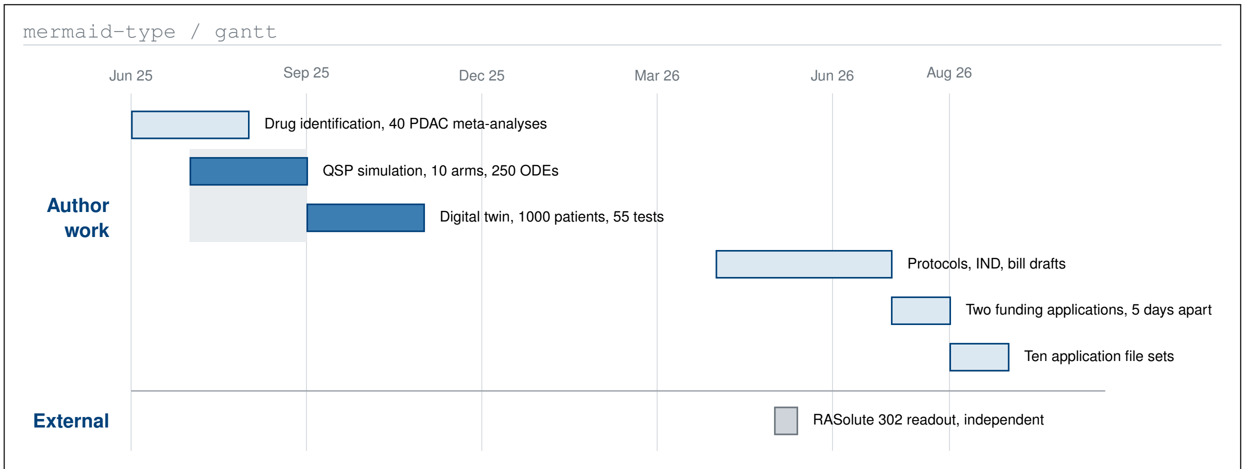
4.3 What Happens If Any One Lands

The join condition is worth stating plainly. Four workstreams run in parallel once any one award lands, and three of them, the interlock rig, the logging schema, and the pharmacy confirmation, complete without reference to which funder paid. Only the site agreement gates what follows, and no funder can supply it.

4.4 The Two Sets Describe One Operation

Set A and Set B are not two programmes. They describe the same hybrid procedure, which carries surgical and medical oncology arms together: a staged eight-arm robotic pancreaticoduodenectomy with perioperative daraxonrasib and an on-premises advisory model, in up to 18 participants under a 3+3 escalation.

What differs is what each set leads with, and that difference is a claim about reviewers rather than about science. A surgical reviewer asked to evaluate a drug schedule will defer to the medical oncology literature; a medical oncology reviewer asked to evaluate an anastomosis will defer to the surgical literature. Writing the same trial twice, each time leading with what the reader can actually assess, is the only honest way to ask both.



The shaded band marks the two months in which two activities ran at once. Bar length is elapsed time, not effort.

Fourteen months of author work against the one external readout. Bar length is elapsed time, not effort, and the two 2025 overlaps are what a single operator could not have run sequentially.

5 Trial Evidence

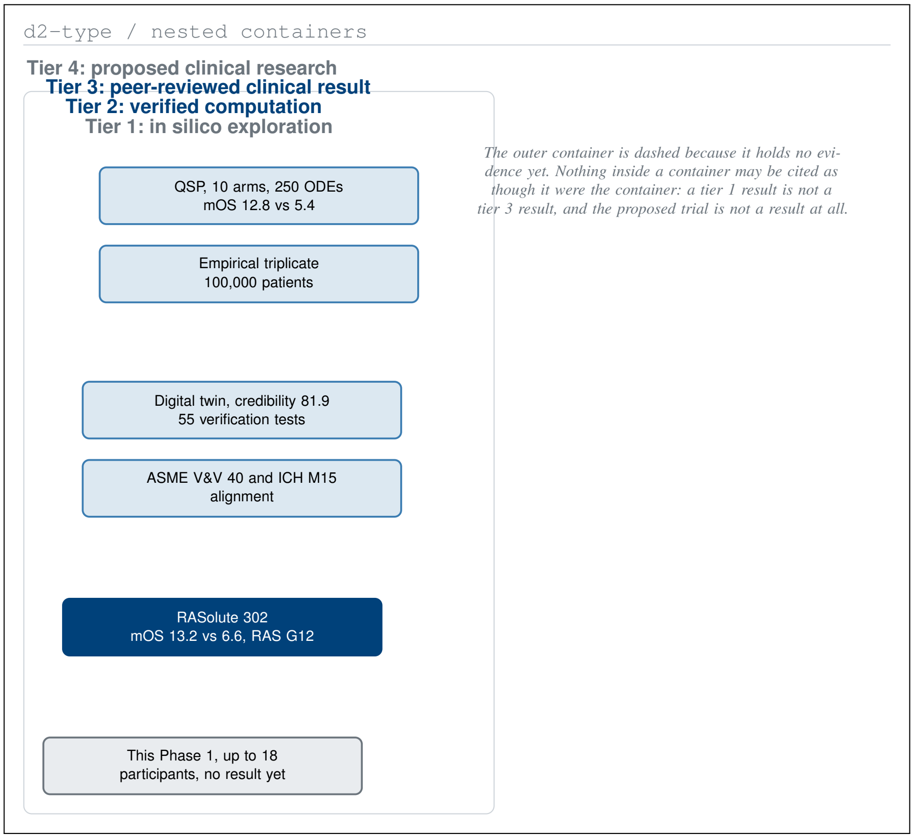
5.1 The Chronology

In June 2025, deep-research meta-analysis over forty pancreatic cancer clinical trial literature areas, totalling more than 400,000 words, produced a ranked shortlist from which daraxonrasib emerged as the leading candidate [14]. In August 2025 a quantitative systems pharmacology simulation of metastatic pancreatic cancer, ten arms and 250 ordinary differential equations per patient, returned a median overall survival of 12.8 months for the daraxonrasib combination against 5.4 months for chemotherapy [4]. In October 2025 a 1000-patient digital twin returned a best median progression-free survival hazard ratio of 0.31 and carried an ASME V&V 40 and ICH M15 aligned credibility score of 81.9 across 55 verification notebook tests [10, 15, 16].

In May 2026, nine months after the QSP run, the RASolute 302 trial reported a median overall survival of 13.2 months with daraxonrasib against 6.6 months with chemotherapy in the RAS G12 population of previously treated metastatic pancreatic cancer [5]. Between June and July 2026 the regulatory package was written, and two complete NIH funding applications were deposited five days apart [17, 18].

5.2 What the Numbers Are, and What They Are Not

Source	Date	Test arm	Control	Stated limitation, in the authors' own words
Empirical triplicate, 100,000 patients	Jul 2025	G3+ AE 8.0%	G3+ AE 25.0%	KRAS G12C log file favours experimental while the KRAS-mutant report favours control; trial-to-trial variability
QSP simulation, 10 arms	Aug 2025	12.8 mo mOS, HR 0.25	5.4 mo mOS	Assumes no acquired resistance and ideal pharmacodynamics; G3+ AE consistently high in all arms
Digital twin, 1000 patients	Oct 2025	12.1 mo mOS, mPFS HR 0.31	not applicable	No patient-specific PK, PD, or tumour growth parameters; simple Emax model, no immune compartments
RASolute 302	May 2026	13.2 mo mOS	6.6 mo mOS	Metastatic and previously treated; silent on the resectable setting



Four evidence tiers drawn as containment rather than as a ladder. Nothing inside a container may be cited as though it were the container, which is the whole of the paper's evidentiary discipline.

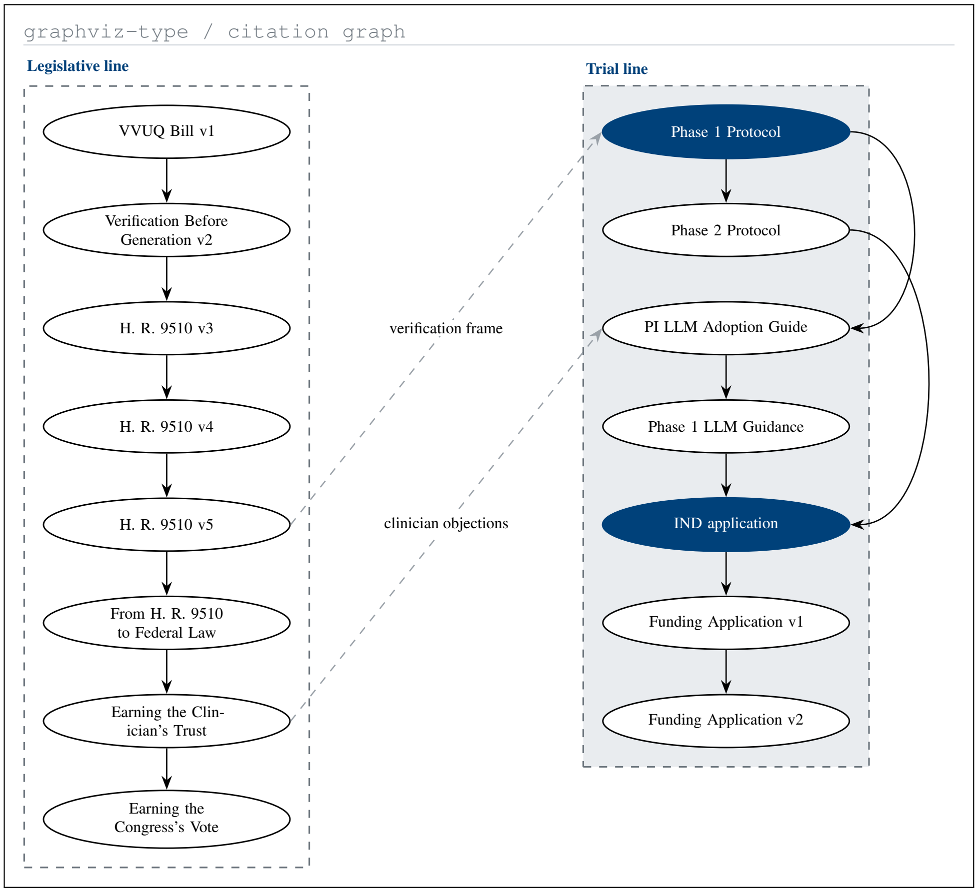
Four sources with the limitation each set of authors stated, carried in the same row as the result. No row can be read without the caveat that qualifies it, which is the condition on citing any of them.

The 2025 simulated survival ratio of 2.4-fold and the 2026 observed ratio of 2.0-fold are close. That proximity is a chronology observation and a hypothesis-supporting one. It is **not** a validation claim, and three differences are material: sample size, 1000 simulated against 241 enrolled; intervention, a daraxonrasib combination against daraxonrasib as a single agent; and molecular selection, KRAS G12C against a primarily G12D and G12V population. Those three sentences appear in every one of the ten applications, in the same paragraph as the numbers, and they appear here for the same reason.

5.3 Cost and Schedule

Work product	Industry cost	Industry schedule	This author
Empirical virtual trial, triplicate	above $120,000 per run	4.5 months	$36,330 estimated, about one month
Ten-arm QSP virtual trial	above $2,000,000	several months to over a year	$36,330 estimated, about one month
Digital-twin simulator platform	$28,000 to $700,000	3 to 16 weeks	$36,330 estimated, about one month

Three work products against their industry cost and schedule benchmarks. The author figures are estimates from the source set and are reported as such, not as audited accounts.

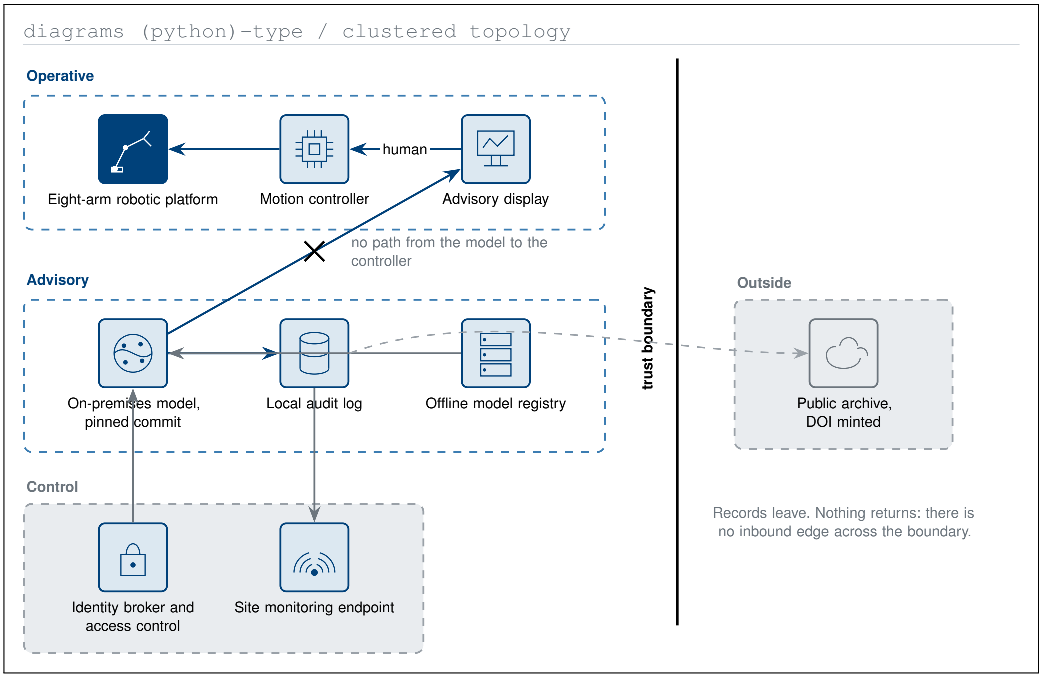


Fourteen deposited works and the citation edges between them. The bill line and the trial line run in parallel and meet only twice, which is why the legislative drafts are not on the trial's path.

5.4 Provenance

Two edges cross the corridor. The verification frame established across the five bill versions enters the Phase 1 protocol, and the clinician objections catalogued in the clinician narrative enter the principal investigator adoption guide. Everything else in the legislative line stays in the legislative line.

That separation matters for a reviewer assessing whether the trial depends on legislation passing. It does not. The bill work supplied a verification vocabulary and a catalogue of objections, and nothing in the protocol, the investigational new drug application, or any of the ten funding applications is contingent on H. R. 9510 being enacted [19, 20].



Nine components, one trust boundary, and one direction of travel. Records leave for the public archive; nothing returns, and no path from the model reaches a motion controller at any point.

6 Physical AI Governance

6.1 Three Boundaries

The model has no path to the arms. Advisory output is rendered to a display. No generated token reaches a motion controller, and no software in the advisory path holds write access to the robot's command channel. The operating surgeon approves every motion [2].

Stopping is bounded and measured. Arm-level stop is specified at 3 milliseconds and system-wide stop at 500 milliseconds. Both are measured on the bench before the first participant is screened and re-measured at every cohort boundary. A specification that is asserted rather than measured is not a boundary; it is a hope.

Every artifact is verified before use. The model runs on premises, pinned to a repository commit, and writes a timestamped record for every recommendation. Conversion to open operation ordered for patient safety is recorded as clinically appropriate; conversion caused by device, advisory, or integration failure is counted separately against the feasibility endpoint [3].

6.2 The Operative Day

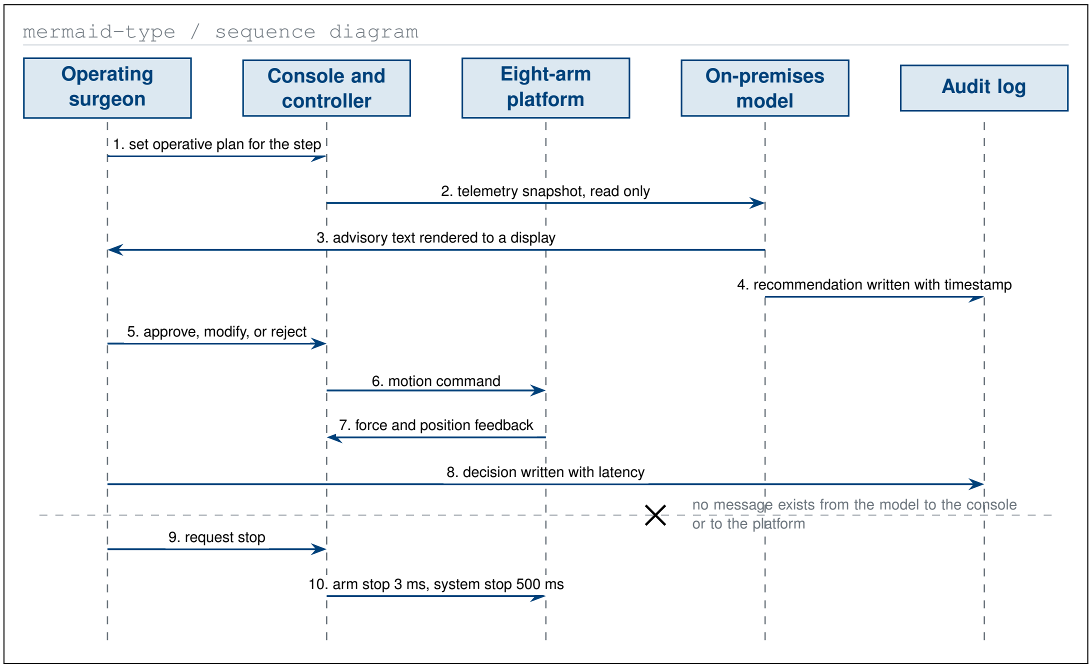
Two figures are needed for one operative day, and neither implies the other. The sequence gives the conversation, message by message, which is what an operating team recognises. The state machine gives the guards, which is what a reviewer checks. A sequence with no guards cannot say what is forbidden; a state machine with no messages cannot say who speaks.

6.3 How a Failure Propagates

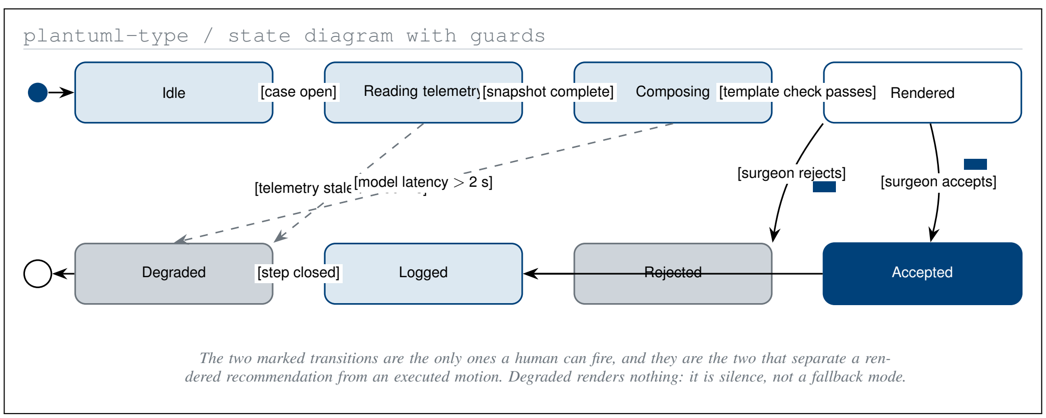
6.4 What Is Released

6.5 Who Produced Which Artifact

The documents this paper summarises were themselves produced by a small set of frontier models with separated roles, and separating them is what makes the review chain meaningful: no model reviews its own output [6].



One operative day as a message sequence. Every arrow reaching the robot leaves a human first, and the two audit writes are the only messages that cross the hospital trust boundary.

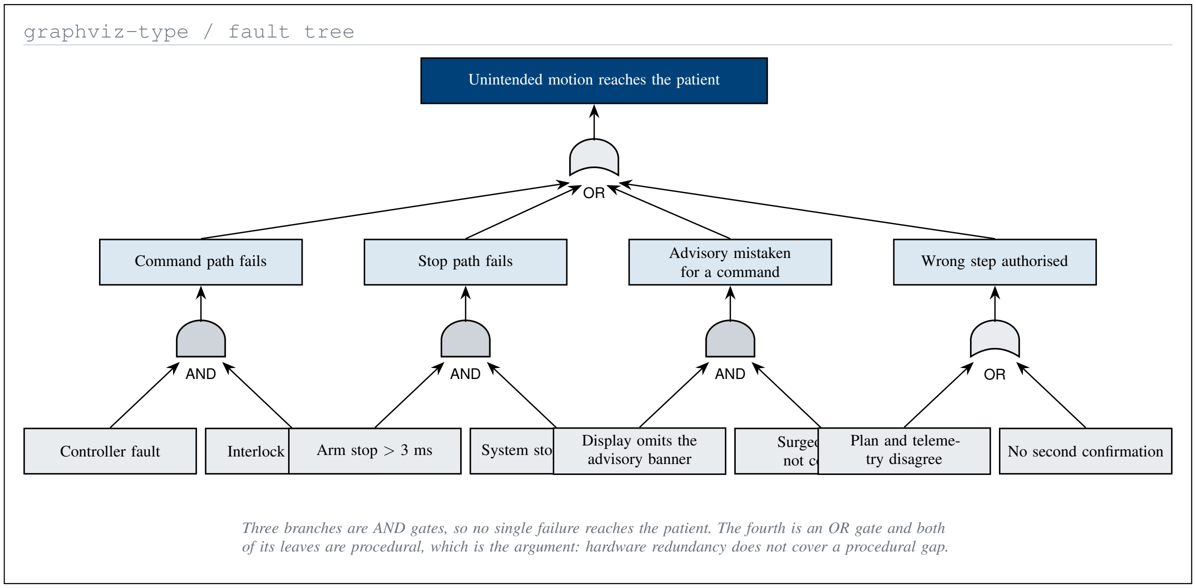


The two marked transitions are the only ones a human can fire, and they are the two that separate a rendered recommendation from an executed motion. Degraded renders nothing: it is silence, not a fallback mode.

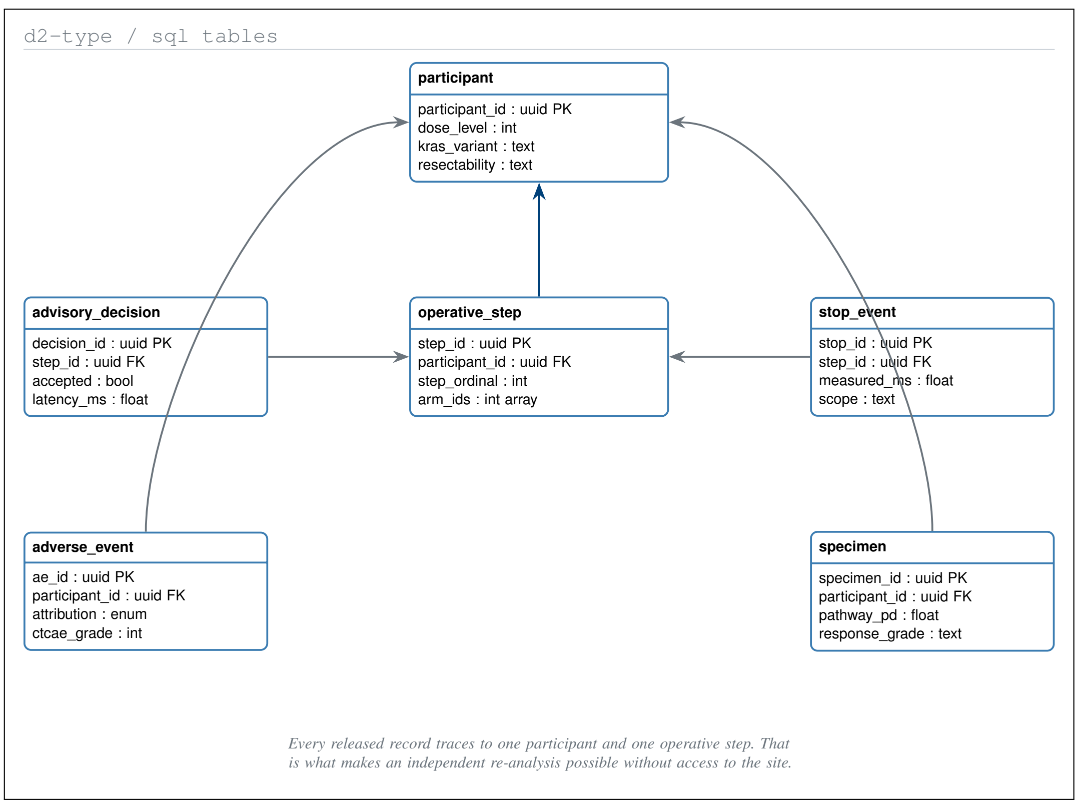
Six advisory states and the guard on every transition. Two guards can only be satisfied by a human action, and they are the two that separate a rendered recommendation from an executed motion.

Model	Role	Artifacts produced
Claude Code	Author	Investigational new drug applications; trial protocols; FDA, ICH and NIH regulation adaptions; legislation revisions; machine-readable diagrams; repository platforms
ChatGPT Codex	First reviewer	Verification, validation and uncertainty quantification of author outputs; meta-analyses of trial papers; external standards; PDF reading and writing
Google Gemini	Second reviewer	Second peer review; accelerated code problem solving; meta-verification of prior stages; administrative automation

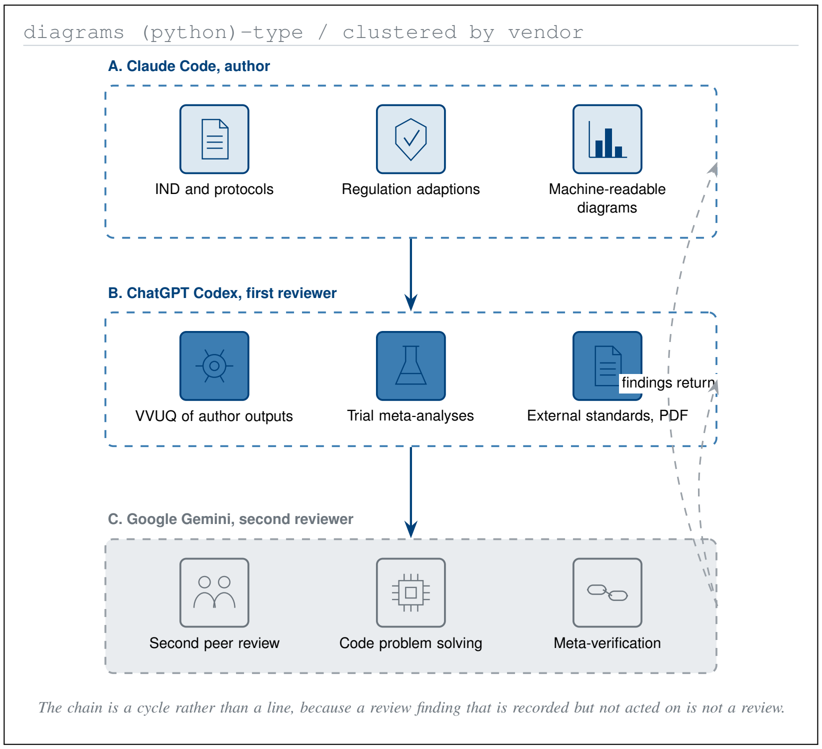
The three frontier-model roles, taken verbatim from the repository's own record. The separation is the point: the author model does not review itself, and the second reviewer checks the first.



One top event and every path to it. Three of the four branches are AND gates, so no single failure reaches the patient, and the one OR branch is the one governed by procedure rather than by hardware.



Six tables and the five keys that join them. Every released record can be traced to one participant and one operative step, which is what makes an independent re-analysis of the cohort possible at all.



Three vendors, nine roles, and the single review chain that connects them. No model reviews its own output, and the chain is drawn as a cycle because the second reviewer's finding returns to the author.

7 UC San Diego Moores Cancer Center

7.1 The Positioning Constraint

UC San Diego Moores Cancer Center is named in all ten applications as the intended partner of choice, approached at the feasibility stage only. It is not described in any of them, or in this paper, as a partner, sponsor, trial site, or endorser, and it will not be so described without written authorization. No agreement of any kind exists. The one application addressed to the institution asks for a 45-minute meeting.

The constraint is not a disclaimer bolted on at the end. It shapes what the applications can claim: a funder reading application 01 is told the site is sought, not secured, and the budget's first-year gate is written as *site executed* rather than as *site confirmed*. An application that overstated the relationship would fail its own go or no-go condition in year one.

This section carries no figure. Application 10 already contains the intake sequence, the role-authority diagram, the feasibility question grid, and the approval dependency, and drawing any of them again here would be duplication rather than a new perspective. The master prompt's requirement that every figure be a unique perspective is a constraint on adding figures, not a quota to fill.

7.2 Three Corrections the Site Research Forced

The most useful thing the site research produced was not a contact list. It was a list of things the applications were getting wrong.

Claim as first drafted	Correction	Where it now appears
Daraxonrasib is first-in-human	It is investigational and already in Phase 3 evaluation. The supportable claim is the first prospective clinical evaluation of the integrated surgical and advisory workflow, subject to FDA and institutional confirmation	A positioning note in nine applications; a pre-send checklist line in all ten
An eight-arm robotic Whipple	No site can assess feasibility or regulatory status without manufacturer, model, cleared configuration, arm count, instruments, and software version. The configuration is specified at the site agreement	Application 10 §1; the positioning note in the other nine
Drug access is available	No drug supply, letter of authorization, or regulatory cross-reference is in place with the agent's developer	Application 06 pre-send instruction; application 10 §1 and back matter

Three corrections the site research forced, and where each now appears. Nine of the ten applications carried the first error until the PART I audit pass, which is recorded here rather than quietly fixed.

The first correction is worth dwelling on because it is the kind of error that survives review. *First-in-human* was true of the integrated workflow and false of the drug, and in a title the two readings are indistinguishable. A reviewer at a drug-development office would have caught it immediately and discounted everything after it. The correction cost two sentences; not making it would have cost the application.

7.3 Two Routes, Sequenced Differently

The repository carries research on two candidate sites, and they are sequenced in opposite directions on purpose.

	UC San Diego	Scripps
Entry point	Pancreatic surgery and GI medical oncology	AI validation and digital-trial methods
First ask	A feasibility meeting on the trial	An independent review of the AI portfolio
The trial appears at	Step 3 of 12	Step 9 of 13
Risk if the route fails	The trial has no site	The AI work has no independent reviewer

The two site routes on four properties. They are sequenced in opposite directions because they are asking different institutions for different things, not because one is a fallback for the other.

The UC San Diego route is trial-first because Moores Cancer Center has the pancreatic surgical volume and the trial operations the study needs, and the question worth asking there is whether the concept is clinically supportable at all. The Scripps route is AI-first because the value it can add is independent evaluation of the verification methods, and asking a methods group to host a surgical trial would waste the introduction.

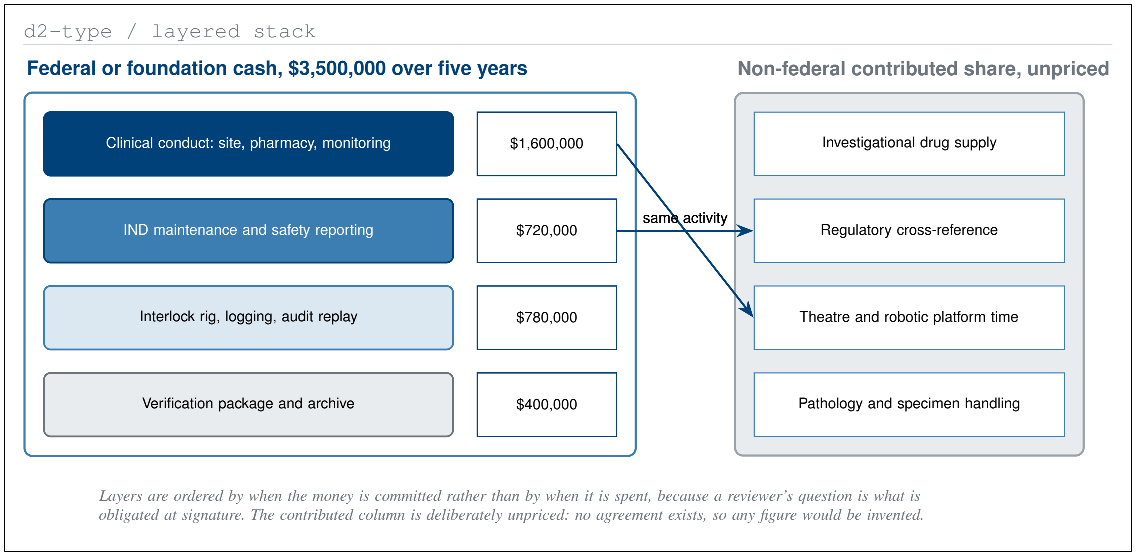
Application 03 uses the contrast directly. Its pathway figure shows the branch taken if the first site agreement does not execute, and that branch is a second qualified site rather than the end of the programme, because the performer, not the site, holds the regulatory package.

7.4 What a Productive Early Outcome Looks Like

The site research sets a deliberately modest bar, and the applications inherit it. A productive outcome is an invitation to present, the identification of a clinical principal investigator, acceptance into an institutional feasibility process, a preliminary budget and accrual estimate, a technical review of the robotic and AI workflow, a defined regulatory and contracting pathway, or support for a grant application. None of those is a trial, and none of them is described as one.

A patient-facing trial proceeds only after the site, the manufacturer, the FDA pathway, the funding, the sponsor responsibilities, the contracts, the insurance, the institutional review board approval, and the operational resources are all documented. That sentence is in the site research, in every application's back matter, and in this paper, and it is the same sentence in all three places.

8 Budget and Leverage



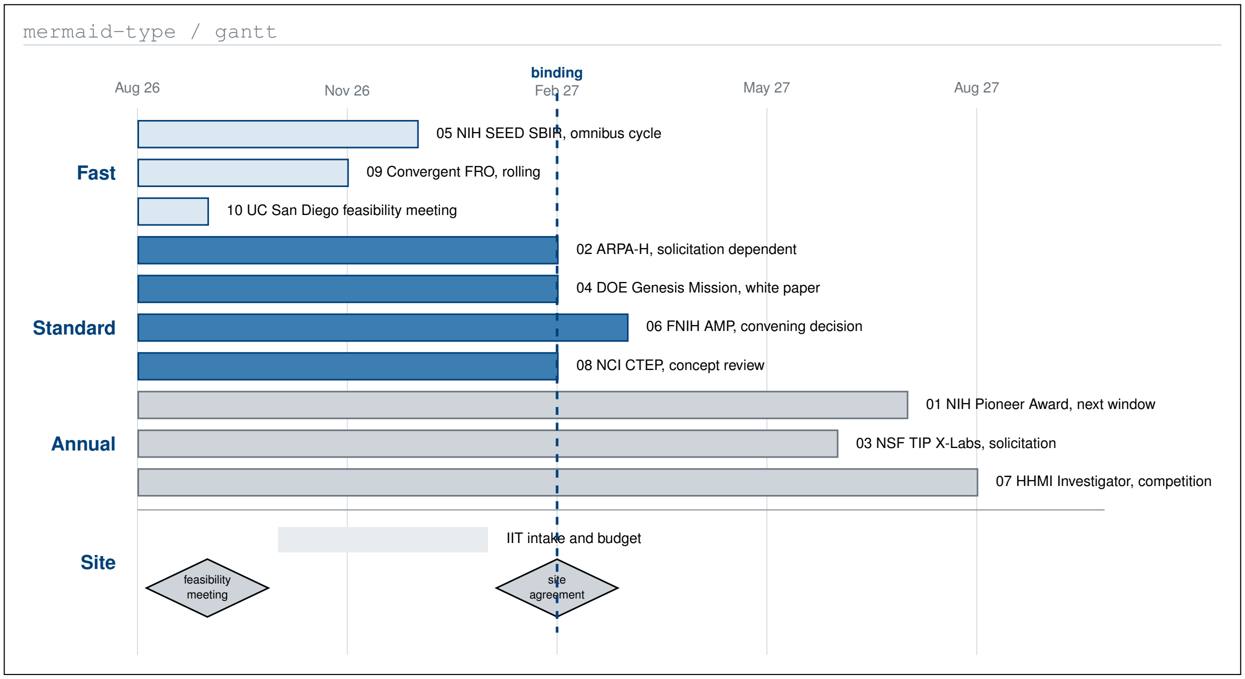
Five years of cash on the left, contributed non-federal value on the right, and the two layers that appear on both sides. The contributed column is what the annex asks agencies to prioritise.

8.1 One Budget, Ten Shapes

The underlying cost is one number: $700,000 per year of direct cost, $3,500,000 over five years, with no cost share and no programme income anticipated [18]. What changes across the ten applications is the shape that number takes, because a mechanism is in part a claim about time.

No	Recipient	Term	Direct ask	Shape, and why the mechanism requires it
01	NIH Pioneer Award	5 years	$3,500,000	Fully funded in year one, as the annex asks for long-duration grants
02	ARPA-H	36 months	$2,100,000	Compressed to three gates, because an ARPA bet should reach a defensible dose or stop
03	NSF TIP X-Labs	5 years	$3,500,000	One public good released per year, so the organization is judged annually
04	DOE Genesis Mission	5 years	$3,500,000	Mission artifacts released in years 1, 2 to 3, and 3 to 5
05	NIH SEED SBIR	9 + 24 months	$1,606,000	Split at a measured gate: $306,000 Phase I, $1,300,000 Phase II
06	FNIH AMP	5 years	$3,500,000 cash	Cash only; drug, theatre, pathology and assay support are contributed
07	HHMI Investigator	7 years	$700,000 per year	Clinical cost only; the balance funds the post-readout pharmacology
08	NCI CTEP	5 years	$3,500,000	Correlative work costed inside, not deferred to a supplement
09	Convergent FRO	5 years	$3,500,000	No year six is budgeted, and year five carries closure costs explicitly
10	UC San Diego Moores	one meeting	none	The only ask that is not money

One underlying budget in ten shapes. The mechanism determines the shape: a gated agency compresses the term, a small-business route splits it at a measured gate, and a time-bound organization budgets its own closure.



Ten submissions sent on one day and ten review clocks that are not the same length. The site milestones beneath them are the binding constraint: no award can start before the site agreement executes.

8.2 Leverage

The annexed FY 2028 memorandum asks agencies to “expand the use of non-Federal cost share, so that Federal dollars draw in and are amplified by private and other non-Federal investment rather than standing alone” [11]. Application 06 is written directly to that instruction: the cash ask covers clinical conduct and regulatory work, and the drug supply, the operating theatre time, the pathology, and the bioanalytical support are contributed rather than purchased.

The contributed column carries no dollar figure, and that is deliberate. No agreement exists with any drug developer or any site, so any valuation would be invented, and an invented cost-share figure is worse than none: it converts a real structural argument into an unverifiable number. The claim made is qualitative and checkable, that four named categories of cost would not appear in the federal request.

8.3 The Binding Constraint Is Not Money

Seven of the ten review clocks close before the projected site-agreement date, and three run past it. The consequence is that the programme’s schedule is governed by the institution rather than by any funder: an award that arrives in month four cannot begin work until month six at the earliest, and the two regulatory activities identified in §2 cannot start until the site is executed regardless of how much money is in hand.

That is why application 10, the one with no financial ask, is the most consequential of the ten. It is the only one addressed to the party that controls the constraint.

9 Build Method

9.1 Two Schedules, Thirteen Sub-Prompts

The whole build runs from one master prompt. That prompt generates thirteen stage sub-prompts in two schedules, and the schedules never share a stage: five drive the ten application file sets, eight drive this paper. Each sub-prompt names its output directory, its commit floor, and the repository files it may read.

Part	No	Stage	Output	Figures
PART I	1	Recipients and templates	Recipient list, cover variants, style and bib contract	none
PART I	2	Set A, surgical	Applications 01 to 05	17
PART I	3	Set B, medical oncology	Applications 06 to 10	17
PART I	4	Email and attachments	Ten ‘.txt‘ files, ten Overleaf zips	none
PART I	5	READMEs and audit	Every ‘funding/**‘ README; the error-fix pass	none
PART II	1	mermaid	Six figure specifications	6
PART II	2	plantuml	Three figure specifications	3
PART II	3	d2	Four figure specifications	4
PART II	4	diagrams (python)	Three figure specifications	3
PART II	5	graphviz	Four figure specifications	4
PART II	6	draft-apply	Skeleton, drafting instructions, 20 figure slots	20 placed
PART II	7	full-apply	Every instruction resolved, every figure drawn	20 drawn
PART II	8	final-apply	Senior-author proof-reading pass	20 polished

The thirteen stage sub-prompts in two schedules. The application stages carry 34 figures across ten attachments; the paper stages carry 20, specified once and drawn twice.

9.2 Why the Diagram Split Is Uneven

An equal split across five platforms would be four figures each. The split used here is 6, 3, 4, 3, 4, and the difference is not arbitrary.

Platform	Count	The question only this vocabulary can state
Mermaid-type	6	What happens next, what decides, how long, and who spoke in what order
PlantUML-type	3	Who is permitted to act, under what guard a state changes, and what runs concurrently
D2-type	4	What contains what, what tabulates against what, and what joins to what
Diagrams (python)-type	3	What runs where, across which trust boundary, and on whose hardware
Graphviz-type	4	What depends on what, how a failure propagates, and what a single record answers

The five platforms, their share of the twenty figures, and the question each is the only one able to state. Forcing an equal split would have moved at least four figures into a vocabulary that cannot carry their subject.

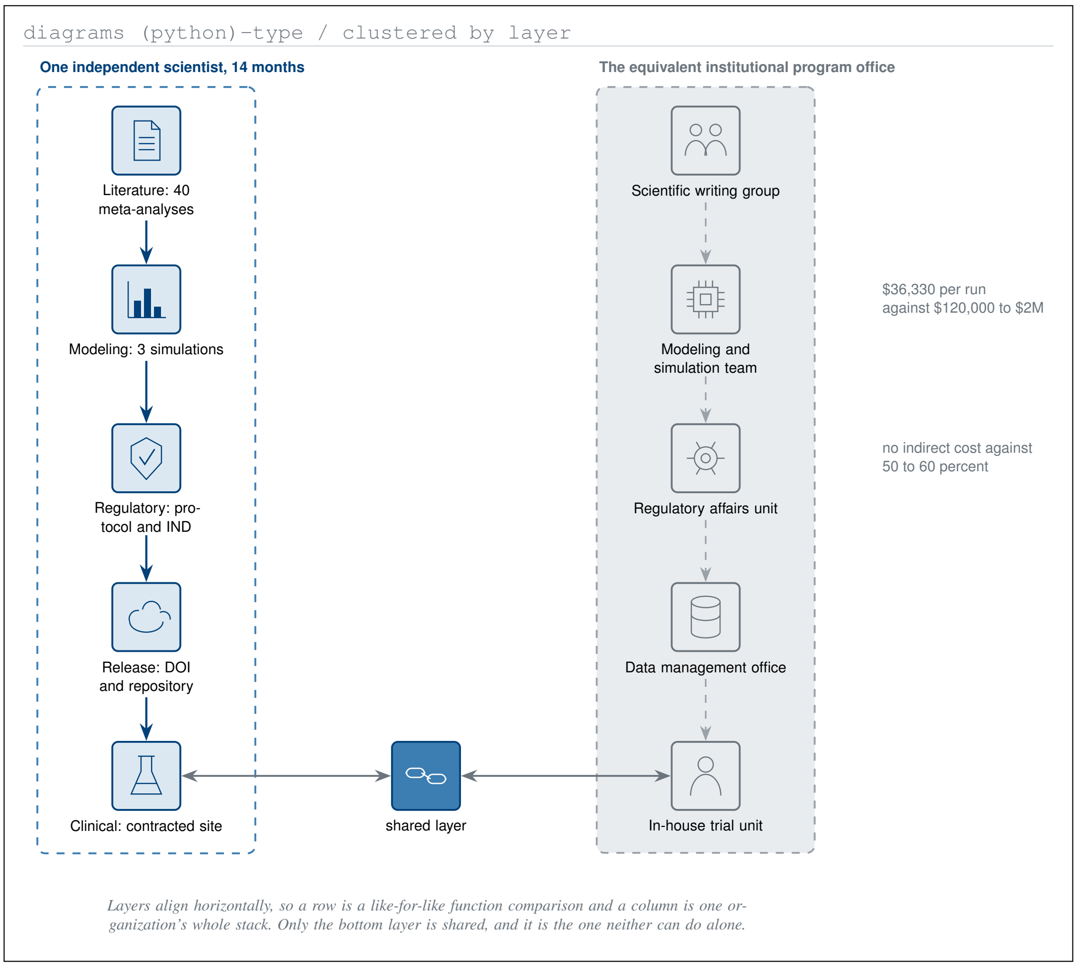
Mermaid takes the largest share because the paper's spine is chronological: a policy document, ten submissions, a fourteen-month record, and one operative day are all sequences. PlantUML takes the smallest because only three subjects in the paper are formal enough to need a guard grammar. Had the split been forced to four each, the authority diagram would have become a flowchart, which cannot draw the absence of a permission; the evidence tiers would have become a ladder, which asserts order where the paper means containment; and the fault tree would have become a dependency graph, which cannot distinguish an AND branch from an OR branch.

9.3 One Operator Against a Program Office

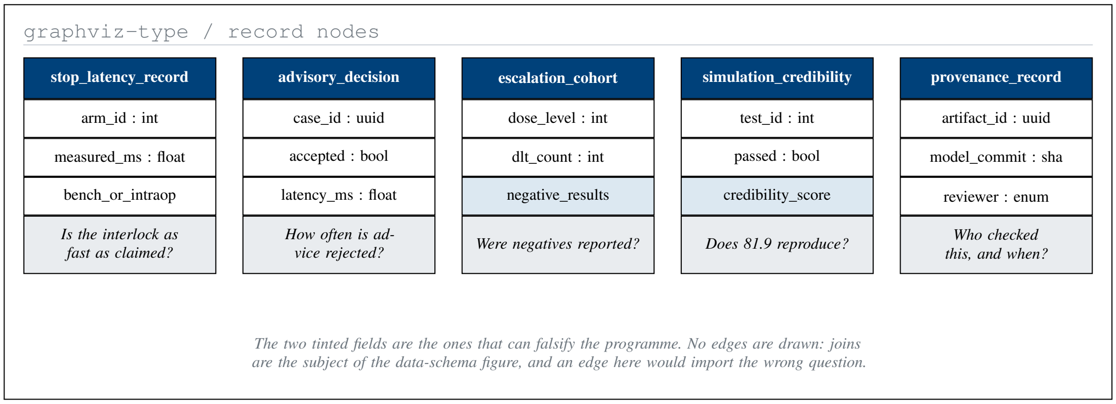
9.4 What Is Reproducible, and What Is Not

Reproducible: the master prompt, the thirteen sub-prompts, the named repository sources, the commit sequence, the figure construction notes, and the compile command. A reader with the repository can rebuild every artifact in it and check every number against the file it came from.

Not reproducible: the judgement exercised at each step. Which recipient qualifies, which vocabulary a figure needs, which sentence in a 123-page policy report is the one worth quoting, and which of the author's own claims is wrong, these are decisions, and a second run would make some of them differently. That is why the author's review is a required stage of this build rather than an optional one, and why the PART I audit pass, which found the first-in-human error in nine of ten applications, is recorded in §7 rather than absorbed silently.



Five layers, executed twice. The left column is one operator with an LLM toolchain and a public repository; the right is the institutional equivalent. Only the bottom layer is common to both.



Five verification artifacts and the reviewer question each field answers. Every field is released with the cohort, and the two marked fields are the ones that can falsify the programme's central claim.

10 Risks and Limits

10.1 What Is Not Claimed

This section is written against the paper rather than for it. Each row below pairs an inference a reader might reasonably draw with what is actually true.

Inference a reader might draw	What is actually true
The simulations validate RASolute 302	They preceded it. The proximity is a chronology observation, and three differences are material: 1000 simulated against 241 enrolled, a combination against a single agent, KRAS G12C against a primarily G12D and G12V population
Daraxonrasib is first-in-human	It is investigational and already in Phase 3 evaluation. The supportable claim concerns the integrated surgical and advisory workflow
A robotic configuration has been specified or cleared	None has. Manufacturer, model, cleared configuration, arm count, instruments and software version are named at the site agreement
Drug supply or a cross-reference exists	Neither exists. No agreement of any kind is in place with the agent's developer
UC San Diego has agreed to something	It has not. One application asks for a 45-minute feasibility meeting
A patient has been treated	None has. The trial does not exist until a site, an IRB, an active IND, monitoring, insurance, and a sponsor of record are all in place
The advisory system holds a regulatory clearance	It holds none, and the applications ask for the study that would generate the evidence a clearance would require

Seven inferences a reader might reasonably draw, and what is actually true in each case. Every row corresponds to a sentence carried in the back matter of all ten applications.

10.2 What Would Falsify the Case

A claim that cannot fail is not a claim. Two released fields can falsify the programme's central assertion, and both are released whatever they show.

The first falsifier is `credibility_score`. The digital-twin work carries a score of 81.9 under ASME V&V 40 and ICH M15 alignment across 55 verification notebook tests [10, 15, 16]. If an independent run of those tests does not reproduce that score, the modelling tier of the evidence stack loses its standing, and with it the arm-selection rationale.

The second is `negative_results`. The applications commit to releasing negative and inconclusive cohorts, and §6's Gold Standard Science mapping makes that a field rather than a promise [21]. A released cohort with that field empty falsifies the programme's claim to Gold Standard alignment directly, whatever the clinical result was.

10.3 Risks the Programme Cannot Retire Alone

Risk	Who can retire it	What the applicant can do meanwhile
No qualified site	An academic cancer center	Keep the regulatory package with the performer, so a second site substitutes without a protocol change
No drug supply or cross-reference	The agent's developer	State plainly in every document that none exists, and design the request so it can be answered in one letter
No sponsor-investigator determination	Institutional contracting and regulatory offices	Ask for the determination before the IRB submission, not after
Robotic configuration not permissible	The site's surgical robotics programme	Leave the configuration unspecified until the site names it
Accrual below plan	The site's referral base	Extend the accrual window rather than widening eligibility

Five risks the programme cannot retire on its own, who can retire each, and the action available meanwhile. None of the five is retired by funding, which is why the funding argument and the feasibility argument are separate.

10.4 The Limit of the Whole Exercise

Ten applications on one piece of science demonstrate that the mechanism is a variable an individual applicant can address. They do not demonstrate that the science is right, that the operation is safe, or that the drug helps in the resectable setting. Those questions are settled by the trial, and the trial does not exist. What this paper reports is the state of the argument as of August 3, 2026, with every number traceable to a named file and every claim paired with the limitation its authors stated.

Back Matter

Abbreviations

Short	Expansion	Short	Expansion
AE	Adverse event	MTD	Maximum tolerated dose
AMP	Accelerating Medicines Partnership	NCI	National Cancer Institute
ARPA-H	Advanced Research Projects Agency for Health	NIH	National Institutes of Health
CTCAE	Common Terminology Criteria for Adverse Events	NSF	National Science Foundation
CTEP	Cancer Therapy Evaluation Program	ODE	Ordinary differential equation
ctDNA	Circulating tumour DNA	OS	Overall survival
DLT	Dose-limiting toxicity	PD	Pharmacodynamics
DOI	Digital object identifier	PDAC	Pancreatic ductal adenocarcinoma
FNIH	Foundation for the National Institutes of Health	PFS	Progression-free survival
FRO	Focused research organization	PK	Pharmacokinetics
HHMI	Howard Hughes Medical Institute	QSP	Quantitative systems pharmacology
IIT	Investigator-initiated trial	RP2D	Recommended Phase 2 dose
IND	Investigational new drug	SAE	Serious adverse event
IRB	Institutional review board	SBIR	Small Business Innovation Research
ISGPS	International Study Group of Pancreatic Surgery	TIP	Technology, Innovation and Partnerships
LLM	Large language model	VVUQ	Verification, validation and uncertainty quantification

Abbreviations used in this paper, set in two column pairs at the body measure. Trial-specific terms follow the definitions in the published Phase 1 protocol rather than being redefined here.

Data and Code Availability

Every work cited to a digital object identifier is openly deposited. The repository is at `https://github.com/kevinkawchak/physical-ai-oncology-trials`, version v4.4.0 [6].

It contains the ten application file sets with their email text and Overleaf bundles, the twenty figure specifications across five machine-readable diagram platforms, the three build stages of this paper, the master prompt and the thirteen sub-prompts, and the TikZ source of every figure printed here. No raster image is used anywhere in this paper or in any of the ten attachments.

Author Contributions and Conflicts

Kevin Kawchak is the sole author. He is chief executive of ChemicalQDevice and holds no equity in Revolution Medicines or in any robotic surgery vendor. No sponsor, contract research organization, institutional review board, regulator, or medical society reviewed this paper or authorised any statement in it. The work was adapted using Claude Code Opus 5.

Citation

Kawchak K. 10 Funding Applications: A Phase 1, First-in-Human, PDAC Clinical Trial Protocol of a LLM-Directed Robotic Whipple with Daraxonrasib (RMC-6236). Zenodo. 2026; 10.5281/zenodo.xxxxxxxx.

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